

Reproduction of facebookresearch/schedule_free: Schedule-Free AdamW on MNIST

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Independent reproduction · Artifact evaluation · Machine learning

Abstract

Ran the repository's MNIST example `examples/mnist/main.py` unmodified at the pinned commit: a small convolutional network trained with Schedule-Free AdamW for 14 epochs (batch 64, lr 0.0025, seed 1), reporting test loss and accuracy after each epoch. One NVIDIA RTX 5060 Ti.

About this record

Title, authors and abstract were provided by the authors and reviewed for completeness, not for scientific soundness. Every number in this record is copied from the signed data of the sealed reports [2], which remain authoritative.

This cover page is not signed. Check it against the SHA-256 listed at <https://kveritas.org/records/2610.00032v1>, and verify each report with its code bundle: `kveritas verify <report.pdf> --bundle <bundle.zip>`.

1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	Schedule-Free AdamW, MNIS...	2026-10-10	1	1	Signed by the K-Veritas server

2 Results

r1: Schedule-Free AdamW, MNIST 14 epochs

Run	Metric	Final value	Step
1	test_loss	0.0281	14
1	test_accuracy	99.35	14

Run	Inline claim	Value	Phase
1	final_test_accuracy	99.35	
1	final_test_loss	0.0281	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	158.4 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	9884f66623d6852083f65f7012c03d36e928182a4b4805520c6f3658a0cff625	1.5 MB
r1.kvbundle.zip	1a8c1c9f0a0f386f33986fbc6bfbe2c2d6d8fdc2fbd7b3dfb565a81a36aed77c	56 KB

5 Links and licence

Paper: <https://arxiv.org/abs/2405.15682>

Reproduction of: https://github.com/facebookresearch/schedule_free/tree/70785b53e778d0e872c0bbb75ff4ee54ee10c291

Record: CC-BY 4.0. Code: Apache-2.0, as declared by the authors. Tags: optimization, schedule-free, adamw, mnist.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (Schedule-Free AdamW, MNIST 14 epochs), kv:2610.00032v1/r1. Data hash 224d6b79199f01e618e7a65ba5597a782d5839827c182ceb2dcc35880e0fbc96.

<https://kveritas.org/records/2610.00032v1/r1.pdf>